Documentation for running project - Turtlebot4 pro - Human Tracker Application Project

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Prerequisites & System Requirements

The following prerequisites are required in order to run the project:

- A custom 3D-printed mounting base for attaching the camera to the upper part of the robot. *Link to the 3D Model*.
- A computer running Ubuntu 24.04 LTS with ROS 2 Jazzy properly installed and configured.
- A Turtlebot 4 Pro platform.
- An OAK-D camera connected to the Turtlebot for real-time visual data acquisition.
- A router capable of providing a stable wireless connection between the computer and the Turtlebot.
    - The router must support a data transfer rate of at least 500 Mbps, due to the continuous transmission of video data from the OAK-D camera to the laptop for real-time processing.
- (Recommended) A computer equipped with a GPU, with the appropriate drivers and CUDA toolkit installed.
    - While not strictly required, GPU acceleration significantly improves the performance and smoothness of the CNN-based processing.

Step 1. Turtlebot 4 pro Setup

1) Connect the turtlebot 4 pro into the dock station (as in *figure. 1*), so it's turn on and battery must be at least 30%

Figure 1. "Connect Turtlebot 4 to dock station"

2) After connecting the Turtlebot to the docking station, it will power on automatically. A wireless network with the SSID: Turtlebot4 will appear on your computer. Connect to this network using the password: Turtlebot4

3) Open a terminal and connect to the Turtlebot via SSH using the following command:

```Shell
ssh ubuntu@10.42.0.1
```

Note: You will be asked for a password, the default password is turtlebot4. When the robot is brand new, the default IP address is 10.42.0.1. If the robot has been previously configured, the current IP address can be found on the Turtlebot's screen (see Figure 2).

Figure 2. "IP Adress"

4) After connecting via SSH, run the following command to configure the Turtlebot network:

```Shell
turtlebot4-setup
```

This command will open the network configuration interface shown in *Figure 3.*

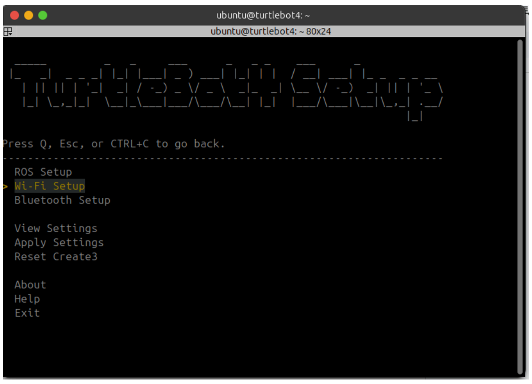

Figure 3. "Connect turtlebot to network"

Select the Wi-Fi Setup option. The network configuration menu shown in *Figure 4* will appear.

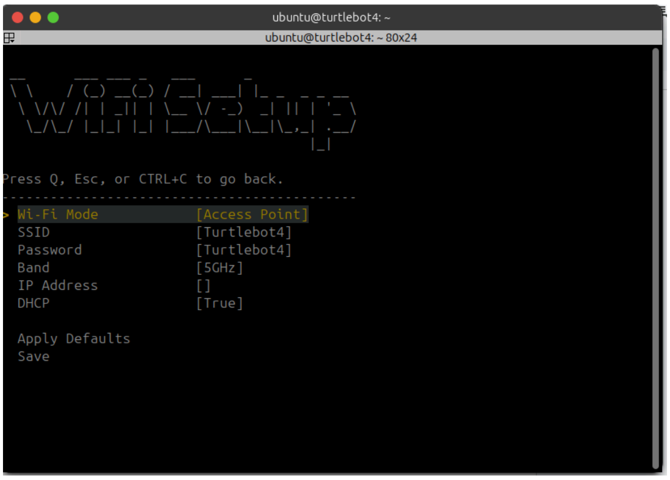

Figure 4. "Network configuration"

Set the network mode to Wi-Fi: Client. Enter the SSID and password of the router specified in the prerequisites. For demonstration purposes, the following example is shown:

- SSID: Turtlebot4Network
- Password: turtlebot4

Note: Do not use these example credentials. Always enter your own router's SSID and password.

The configuration process is illustrated in Figure 5.

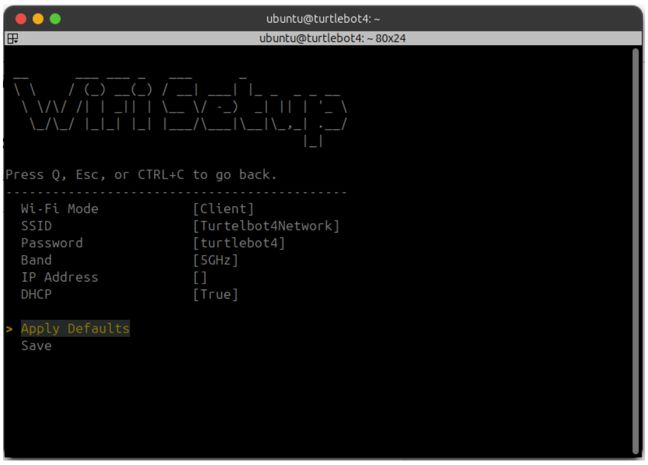

Figure 5. "Network configuration modification"

Click Save to apply the changes. When prompted, enter the user password: turtlebot4.

Finally, click Apply Settings, as shown in Figure 6.

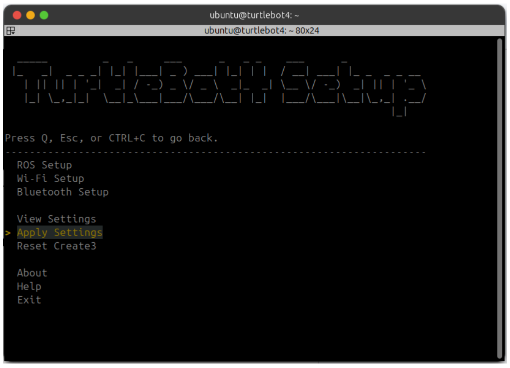

Figure 6. "Apply Settings"

Confirm the changes by clicking Yes, as shown in Figure 7.

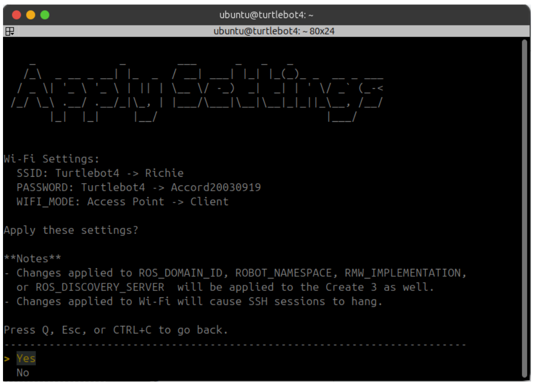

Figure 7. "Save Settings"

After applying the configuration, a message with warnings might appear in the terminal, as shown in Figure 8.

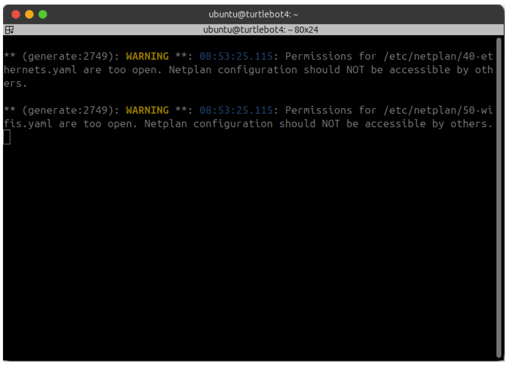

Figure 8. "Finish network configurations"

At this point, the Turtlebot will attempt to connect to the configured router. Once connected, the robot will be ready to run the project. If the connection is not established immediately, power off the Turtlebot by pressing the large round button. Then, place it back on the docking station to power it on again. After rebooting, the robot should automatically connect to the router.

Note: Once connected to the router, the Turtlebot's IP address will change.

Step 2. Setup your Computer

1) Ensure that your computer has an active internet connection, then clone the following GitHub repository:

```shell
git clone https://github.com/Ricardo0919/turtlebot-4-pro-trackerProject.git
```

Note: For the next steps, ensure that ROS 2 Jazzy and rqt_image_view are installed, pip dependency of python as they are required for running the project and visualizing the video stream.

- Installation of Ros2 Jazzy Ubuntu 24.04

For rqt_image_view

```shell
sudo apt update
sudo apt install ros-jazzy-rqt -y
```

2) Next, install the Python dependencies listed in requirements.txt. Navigate to the root directory of the cloned repository and run the following command:

```python
python3 -m pip install --break-system-packages -r requirements.txt
```

3) After installing the requirements, navigate to the tracker_ws folder *(workspace)* and run the following commands:

```shell
source /opt/ros/jazzy/setup.bash
colcon build
```

Note: After completing the build process, you should see an output similar to the one shown in Figure 9. Do not close this terminal window, as it will be used later.

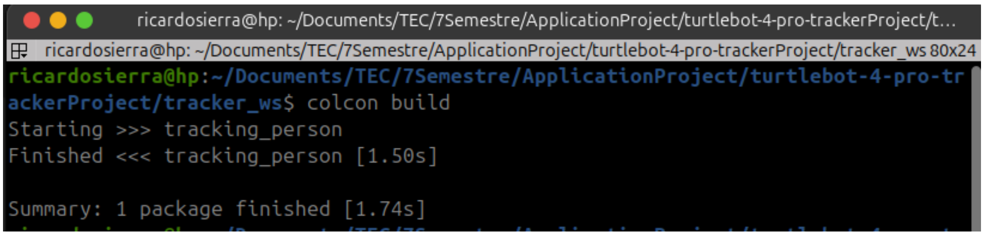

Figure 9. "Output Colcon Build"

Step 3. Run the project

1) Ensure that both the TurtleBot and your computer are connected to the same router network. In this example, the network SSID was Turtlebot4Network.
2) Open a new terminal window and connect to the TurtleBot via SSH using the following command. Replace the IP address with your own:

```shell
ssh ubuntu@192.160.0.6
```

Note: You will be asked for a password, the default password is turtlebot4.

3) In the terminal where you are logged into the TurtleBot, run the following command to enable and use the robot's camera:

```shell
ros2 launch turtlebot4_bringup oakd.launch.py
```

After the command finishes, the message "Loaded node '/oakd' in container 'oakd_container'" will appear in the terminal. Once this message is displayed, press Ctrl + C.

4) Now, in the same terminal where colcon build was previously executed, run the following commands:

```shell
source install/setup.bash
ros2 launch tracking_person Tracker_launch.py
```

After running the commands, two rqt_image_view windows should appear, as shown in *Figure 10.*

- These windows allow you to directly visualize the image streams being processed.
- The image published on the topic /oakd/rgb/preview/image_raw corresponds to the raw image directly captured from the camera.

The image published on the topic /tracking_person/annotated corresponds to the processed image used for person tracking.

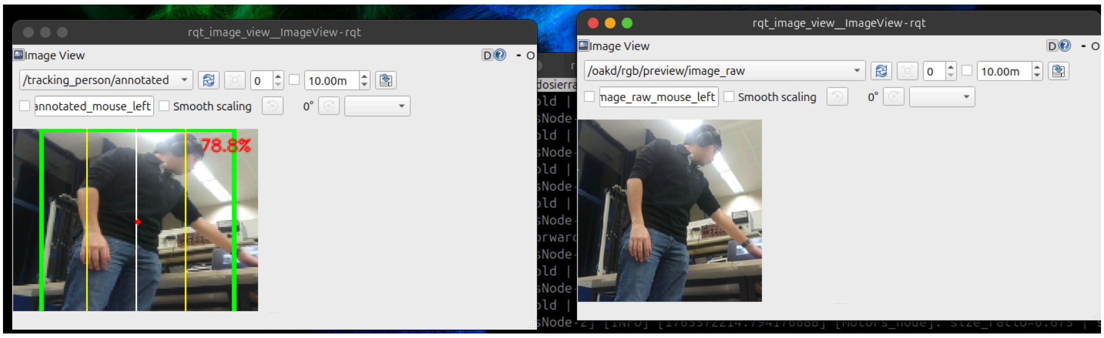

Figure 10. "Visualization of camera and tracking output"

5) If you have reached this point, you have successfully set up and executed the project.